Comparing Lexical, Static-Embedding, and Contextual Drift Signals in Brazilian Portuguese Political Discourse

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Abstract. *The shift in political discourse often reflects deeper societal changes, but quantifying how specific terms evolve semantically over decades remains a significant challenge for computational linguistics. In this study, we propose a cost-aware comparative framework and use it to compare three families of diachronic drift signals in Brazilian Portuguese political discourse, namely a lexical-profile baseline based on TF-IDF, a static embedding pipeline based on slice-specific Word2Vec models with Orthogonal Procrustes alignment, and a contextual analysis based on a Portuguese BERT checkpoint. Using a BrPoliCorpus subset (428k speeches from 2000–2023), to make the methods directly comparable, we build a shared panel of 55 lemmas containing method-specific drift candidates, stable controls, and theory seeds. The resulting comparison shows strong disagreement between the lighter baselines, with Spearman correlation of* -0.54 *and zero overlap in their top 15 terms, while contextual BERT remains moderately closer to Word2Vec than to TF-IDF. A filtered contextual panel reduces stable-control leakage and highlights terms such as "bloqueio", "salário", "intervenção", and "eleição". Rather than treating one detector as externally validated ground truth, we frame the study as an exploratory comparison of complementary drift sensitivities, interpretability, and computational cost in Brazilian Portuguese political discourse. The framework itself is model- and language-agnostic, offering a reusable blueprint for multi-method drift studies.*

1. Introduction

The language of politics reflects historical change, as shifts in ideological agendas, coalition structures, and policy framing become observable in patterns of lexical choice and word usage over time. However, manually analyzing large-scale and evolving patterns in political discourse is challenging. Diachronic natural language processing (NLP) methods offer promising avenues for analysis. Among these, distributional models can capture some of these movements at scale, but different modeling choices foreground different aspects of change [Hamilton et al. 2016, Kutuzov et al. 2018, Tahmasebi et al. 2021]. In political corpora the situation is especially complex, since a term may drift because of a new policy referent, a rhetorical reframing, a shift in collocational profile, or a reorganization of discourse coalitions, and none of these mechanisms reduces neatly to a single lexical sense change. To structure this comparison, we introduce a cost-aware comparative framework that stages detection methods by computational expense and channels their outputs through a shared comparison panel for cross-method agreement analysis.

This complexity carries over to evaluation, where shared-task benchmarks have improved comparability in lexical semantic change detection [Schlechtweg et al. 2020] but curated external labels remain scarce, especially outside the few languages and

corpora covered by standard tasks. Distributional pipelines can also react to temporal artifacts, alignment noise, and other confounds when outputs are read too literally [Dubossarsky et al. 2019]. Without externally validated ground truth, the central question becomes not which detector is "correct" but what each detector appears to capture when applied to the same political timeline.

This question is timely for Brazilian Portuguese. Large-scale political corpora such as BrPoliCorpus [Lima-Lopes 2025] are only now becoming available, yet comparative drift studies that place multiple detection families side by side on the same Portuguese political material remain rare. We instantiate the framework on the floor subset of BrPoliCorpus, which covers 24 yearly slices of Chamber speeches from 2000 to 2023, comparing a TF-IDF lexical-profile baseline, a slice-specific aligned Word2Vec pipeline, and a contextual BERT stage, all scored on a shared comparison panel of 55 lemmas. The paper addresses the following questions:

1. How strongly do the methods agree on the same lemmas?
2. Which candidates remain method-specific?
3. Does the higher-cost contextual stage add enough interpretive value to justify its expense?

Beyond addressing these questions for Portuguese political discourse, where published multi-method drift analyses remain much scarcer than in the better-studied English-centered literature, this work contributes a reusable framework design that can be instantiated with alternative models, corpora, and languages.

2. Related Work

Distributional semantic change detection has evolved from co-occurrence comparisons to aligned static embeddings and then to contextual representations [Kutuzov et al. 2018, Tahmasebi et al. 2021]. Among static approaches, slice-specific Word2Vec models aligned with Orthogonal Procrustes have been shown to recover known historical shifts and reveal statistical regularities of change [Hamilton et al. 2016]. Such embeddings remain attractive because they are efficient and produce interpretable neighborhood movements, but they compress all usages of a word into a single vector, blurring cases in which older and newer usages coexist.

Evaluation remains the harder problem. Standardized shared tasks have improved comparability [Schlechtweg et al. 2020], yet many real-world studies still operate without external semantic ground truth, and apparent change can emerge from temporal referencing, alignment noise, or genre shifts rather than from semantic drift alone [Dubossarsky et al. 2019]. This concern motivates the stable controls and agreement analysis in the present work.

Contextual models offer a partial response to the aggregation problem. BERT-style representations encode token-level usages and can surface variation that static spaces smooth over [Giulianelli et al. 2020, Devlin et al. 2019]. Scalable contextual methods that improve interpretability have also been proposed [Montariol et al. 2021], and pre-trained Portuguese checkpoints such as BERTimbau [Souza et al. 2020] have made contextual analysis feasible for Portuguese NLP. We place contextual BERT beside TF-IDF and aligned Word2Vec on the same political panel so that agreement, disagreement, and computational tradeoffs can be examined directly.

Diachronic analysis of political language has gained traction through parliamentary corpora. The study of [Hofmann et al. 2020] compared lexical usage in Austrian political discourse across time, but comparable multi-method studies on Brazilian Portuguese political speech remain scarce. Existing Portuguese diachronic resources [Osório and Lopes Cardoso 2025] have focused primarily on historical and literary material rather than political discourse, leaving a gap that BrPoliCorpus [Lima-Lopes 2025] now fills with the scale needed for controlled comparative work.

3. Comparative Drift Framework

The framework organizes lexical semantic change detection into three computational tiers arranged by increasing cost, each targeting a distinct facet of diachronic drift. The first tier, a **lexical-statistical baseline**, operates on surface-level frequency or weighting counts and requires no model training, capturing shifts in lexical salience at negligible computational expense. The second tier, a **static-embedding method**, maps each word type to a single dense vector per temporal slice and aligns slice-specific models into a common geometric space so that drift becomes a measurable displacement; training cost is moderate and results are interpretable through nearest-neighbor inspection. The third tier, a **contextual-embedding method**, relies on a pretrained language model that produces a distinct vector for every token occurrence, which means this tier can in principle detect sense-level divergence that the cheaper stages miss, though it is the most expensive and the hardest to interpret.

The key design principle is *staged cost awareness*. The two cheaper tiers screen the full scored vocabulary first, and their method-led candidates feed forward into the expensive contextual stage, which avoids running heavy inference over every eligible lemma. Cross-method comparison then requires a fixed evaluation set that prevents each method from being assessed only on the terms it would naturally favour. The framework therefore prescribes a *shared comparison panel* assembled from four sources: (i) high-drift candidates nominated by the lexical-statistical baseline, (ii) high-drift candidates nominated by the static-embedding method, (iii) low-drift stable controls selected jointly by both cheaper tiers, and (iv) theory-driven seed terms drawn from domain expertise or prior work. Because all three tiers score exactly the same candidate set, any disagreement reflects genuine differences in what each method measures rather than divergent vocabularies.

On the shared panel the framework computes an *agreement analysis layer* consisting of pairwise rank correlations, top-k overlap curves, and bucket-level rank distributions that separate drift candidates from stable controls. These diagnostics reveal where the tiers converge, where they diverge, and whether the more expensive contextual stage adds enough interpretive value to justify its cost; the analysis layer itself is agnostic to the specific models plugged into each tier.

The framework assumes a diachronic corpus partitioned into temporal slices, where each slice must contain enough documents and tokens to support stable frequency estimates and distributional model training, and a preprocessing step normalizes surface forms to a shared lemma representation so that all three tiers operate on the same vocabulary. Not every word form is suitable for drift scoring, so the framework applies three joint filters before any method runs: a minimum per-slice frequency to guarantee sufficient token mass for embedding training, a document-dispersion threshold to prevent a single source

from inflating apparent salience, and a minimum slice-presence rate to restrict attention to genuinely diachronic terms rather than episodic ones. Drift and stable-control candidates may be further restricted to dominant part-of-speech classes.

Figure 1 summarizes the staged pipeline. Although we present a particular instantiation in Section 4, the framework itself is model- and language-agnostic, and any combination of methods that respects the three-tier cost ordering can be substituted.

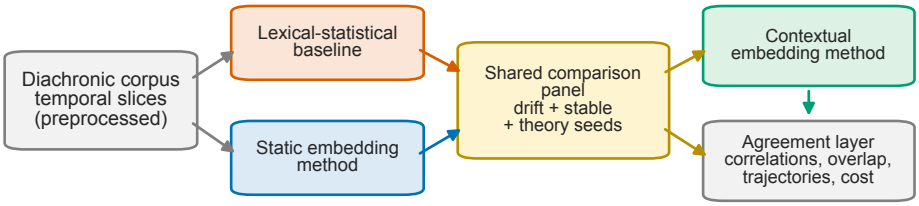

Figure 1. Staged comparative framework for lexical semantic change detection. Two cheaper tiers screen the scored vocabulary and nominate drift candidates; a shared comparison panel combines those candidates with stable controls and theory seeds before the expensive contextual tier and agreement analysis.

4. Methodology

We instantiate the comparative framework of Section 3 on Brazilian Portuguese parliamentary discourse. Each tier is realized with a specific, well-established model: TF-IDF for the lexical-statistical baseline, slice-aligned Word2Vec with Orthogonal Procrustes for the static-embedding tier, and BERTimbau Large for the contextual tier. The individual techniques are grounded in the literature [Hamilton et al. 2016, Giulianelli et al. 2020]; our contribution is their orchestration through the staged pipeline and shared panel design.

4.1. Corpus and Preprocessing

The experiments use the floor subset of BrPoliCorpus [Lima-Lopes 2025], which contains Brazilian Chamber floor speeches from 2000 to 2023 organized into 24 yearly slices. A growing number of diachronic Portuguese resources are now available [Osório and Lopes Cardoso 2025], ranging from syntactically annotated literary collections such as the Tycho Brahe Parsed Corpus [Galves and Faria 2017] to large multi-genre repositories such as the Corpus do Português [Davies and Ferreira 2006]. These resources target historical-linguistic research across broad periods rather than domain-specific political analysis with continuous yearly slices. Within BrPoliCorpus itself, the inaugural-speech subset reaches back to 1889 but contains only 35 documents, too few for stable distributional estimation. The floor subset offers the largest contiguous yearly coverage (428,366 speeches across 24 slices), a uniform parliamentary register, and exact date metadata, making it the most suitable segment for a year-by-year comparative drift study in the political domain.

Raw speeches are tokenized and tagged with `spaCy` (`pt_core_news_sm`). We retain nouns, verbs, and adjectives, exclude proper nouns, remove stopwords, punctuation,

and numeric tokens, and keep only tokens with at least two characters. All text is lowercased with accents preserved. Each token is then mapped to its lemma form, producing the `content_lemma` representation used by all three methods. After preprocessing, the corpus retains 428,366 speeches and 63,036,642 tokens across 225,484 unique lemmas. Table 1 summarizes the corpus totals and the quartile structure of yearly variation.

A lemma enters the scored vocabulary only if it appears at least 50 times per slice, in at least 5 documents per slice, and in at least 80% of the 24 slices. These thresholds were fixed as practical corpus-stability defaults rather than tuned to optimize any particular method. The frequency floor ensures sufficient token mass for reliable embedding training in each yearly slice; the document-dispersion requirement prevents a single speaker or parliamentary session from inflating a lemma's apparent salience; and the slice-presence constraint restricts attention to lemmas that are genuinely diachronic rather than episodic. Drift and stable-control candidates are further restricted to nouns and adjectives, and a small set of procedural high-frequency terms is excluded. These filters reduce the scored vocabulary to 3,221 lemmas for the two cheaper methods.

Yearly document counts range from 3,041 to 26,868 (median 16,752) and retained token volume from 630K to 4.2M (median 2.6M). Document volume is high throughout, but token and vocabulary variation across slices motivates the frequency and slice-presence controls described above.

Table 1. Corpus summary for the BrPoliCorpus floor subset. Yearly slice statistics are reported as the 25th percentile (Q1), median, and 75th percentile (Q3) across the 24 yearly files; token and character counts are shown in millions.

Period	Slices	Measure	Corpus total	Yearly slices		
				Q1	Median	Q3
2000–2023	24	Speeches	428,366	14,648	16,752	21,987
		Tokens	63.0M	2.00	2.62	3.45
		Characters	538.5M	16.9	22.3	29.7
		Lemmas	225,484	39,456	50,269	58,637

4.2. TF-IDF Baseline

For each eligible lemma we compute a TF-IDF vector across yearly slices using smoothed inverse-document frequency. Drift is measured as the mean absolute change in TF-IDF weight between adjacent slices. This captures shifts in lexical salience, reflecting how central a term becomes to each year's agenda, without relying on distributional neighborhoods.

4.3. Word2Vec Baseline

We train independent skip-gram models per yearly slice (300 dimensions, window 5, 5 negative samples, minimum count 5, 5 epochs, 2 replicates with different seeds) and align them into a common space with Orthogonal Procrustes over the top 20,000 anchor words [Hamilton et al. 2016]. Drift is the mean cosine distance between a lemma's aligned vectors in consecutive slices, averaged over replicates. This captures persistent neighborhood reorganization.

4.4. Contextual BERT

We embed sampled in-context occurrences with a Portuguese BERT checkpoint (BERTimbau Large [Souza et al. 2020]), extracting hidden-state layers -1 and -4. For each lemma and slice, up to 64 occurrences are sampled within a 48-token context window; the contextual drift score is the mean cosine distance between slice-level centroid embeddings in consecutive slices. The agreement layer keeps layer -1 as the preferred ranking and uses layer -4 as a robustness view.

4.5. Shared Comparison Panel

Following the panel design prescribed by the framework (Section 3), we construct a shared panel of 55 lemmas: 15 Word2Vec-led drift candidates, 15 TF-IDF-led drift candidates, 20 stable controls (lemmas ranked lowest by both cheaper methods), and 5 theory seeds drawn from prior work on Brazilian political vocabulary (*democracia*, *corrupção*, *reforma*, *economia*, *liberdade*). All three methods observe the same candidate set, so disagreement reflects properties of the signal rather than divergent vocabularies. The agreement layer derives pairwise rank correlations, top-k overlaps, stable-control leakage diagnostics, and a filtered contextual ranking.

Figure 2 visualizes the pairwise agreement structure, while Figure 3 summarizes overlap and rank-distribution statistics.

5. Results

The two lighter baselines disagree strongly even after they are constrained to the same candidate universe. On the shared panel, Word2Vec and TF-IDF reach Spearman $\rho = -0.54$ ($p = 2.1 \times 10^{-5}$) and share zero terms in their top 15 rankings. The preferred contextual layer partly mediates this split without collapsing onto either baseline. Its correlation with Word2Vec is $\rho = 0.21$ ($p = 0.128$) and with TF-IDF $\rho = 0.12$ ($p = 0.365$). These positive associations are directionally informative but not strong enough to justify treating contextual BERT as a substitute for either cheaper method. Figure 2 visualizes this pairwise rank structure. Each panel plots the drift scores of the 55 lemmas under two methods, with Spearman ρ and p-values annotated. The Word2Vec vs. TF-IDF scatter reveals the strong negative trend where terms ranked highly by one baseline tend to be ranked low by the other, confirming that they capture fundamentally different facets of change. The BERT vs. Word2Vec and BERT vs. TF-IDF panels show that contextual BERT draws partial support from both camps without collapsing onto either, while the BERT layer comparison confirms that the two extraction layers agree closely with each other.

Overlap statistics and bucket-level rank distributions prove more informative than raw top-k lists alone. In the preferred BERT layer, the top-15 overlap is 7 terms with Word2Vec and 6 terms with TF-IDF, whereas the baseline-to-baseline overlap remains zero. Figure 3 also shows that selected drift terms are ranked significantly above stable controls by Word2Vec and TF-IDF ($p < .001$ in both cases), while contextual BERT still separates the groups but with weaker margins ($p = .008$). This pattern is consistent with the filtered contextual panel, where BERT provides useful adjudication but the raw contextual ranking still leaks some stable controls and should not be treated as an unrestricted final list. The two contextual extraction layers nevertheless agree strongly with each other (Spearman

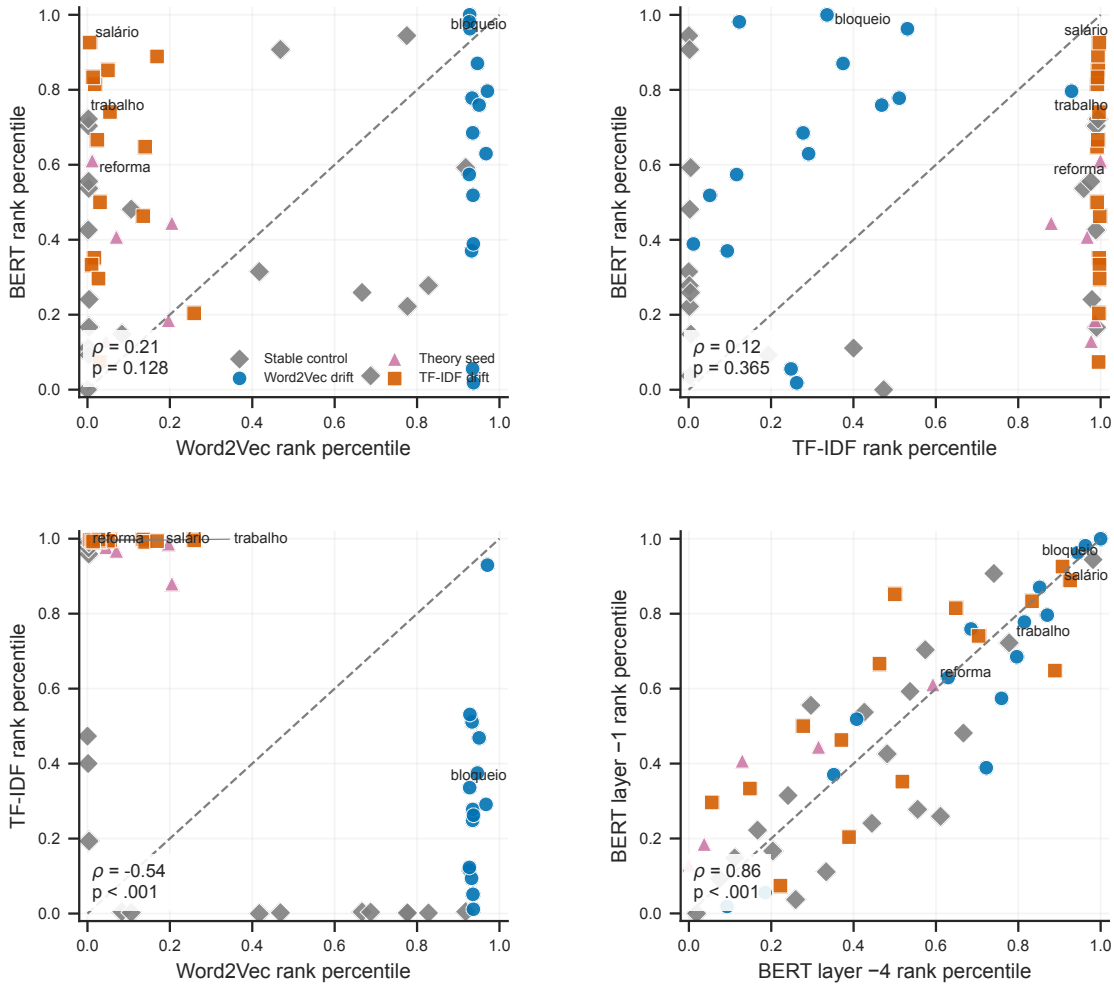

Figure 2. Pairwise rank agreement on the shared comparison panel across BERT vs. Word2Vec (top left), BERT vs. TF-IDF (top right), Word2Vec vs. TF-IDF (bottom left), and BERT layers -4 vs. -1 (bottom right). Annotations report Spearman ρ and two-sided p-values; legend in the top-left panel applies throughout.

$= 0.858$), so the broad comparative picture does not depend on a single extraction depth. This high inter-layer agreement accords with findings that mid-to-upper layers carry most of the semantic change signal, while the final layer increasingly encodes surface-form information that can bias change scores [Laicher et al. 2021, Cassotti et al. 2024]; the stability between layers -1 and -4 in our setting suggests that orthographic bias does not yet dominate, though it does not rule it out.

As a robustness check, we recomputed contextual scores using average pairwise distance (APD) [Giulianelli et al. 2020] instead of the centroid-based metric. APD operates on all individual occurrence embeddings rather than on slice-level centroids, giving equal influence to each usage regardless of embedding magnitude. On the preferred layer, APD and centroid rankings correlate at Spearman $\rho = 0.79$ ($p < 10^{-12}$), with 7 of 15 top terms shared. The two metrics foreground different properties: APD promotes high-frequency polysemous terms whose embedding clouds are broad (e.g. *trabalho*, *público*),

while the centroid metric favours terms with coherent directional shifts (*bloqueio*, *salário*). Notably, the centroid metric separates method-nominated drift candidates from stable controls more cleanly than APD ($p = .008$ vs. $p = .081$, Mann–Whitney). The strong overall correlation confirms that the comparative conclusions are robust to the choice of contextual aggregation, while the separation difference suggests that centroid distance is a less noisy signal for this panel-based comparison.

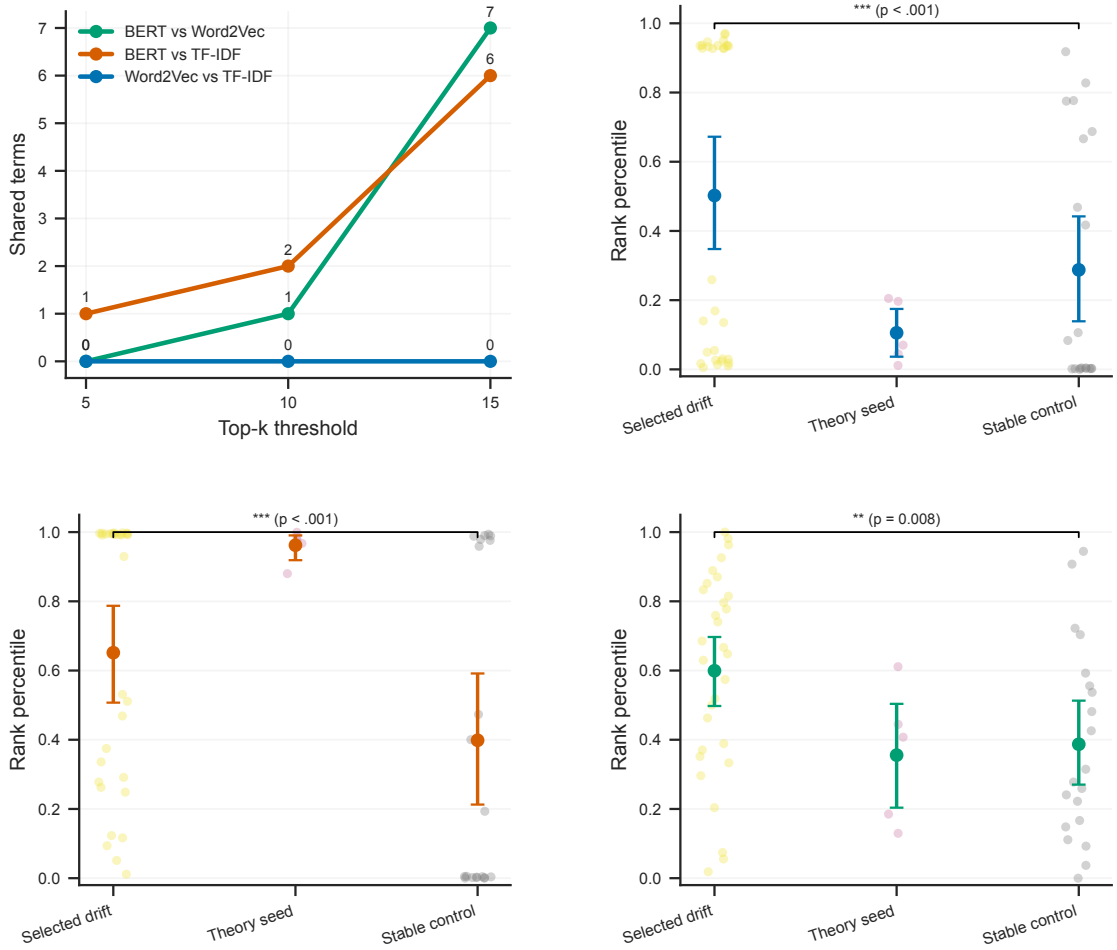

Figure 3. Overlap and rank-distribution summaries on the shared comparison panel. Top left shows top-k overlap between methods; top right through bottom right show rank-percentile distributions by bucket for Word2Vec, TF-IDF, and BERT respectively. Points are individual terms; large markers are bootstrap means with 95% confidence intervals; brackets report one-sided Mann–Whitney tests for drift terms vs. stable controls.

6. Discussion

Representative term trajectories show that the methods often disagree about *when* a term moves, not only whether it moves at all. Figure 4 illustrates four cases: a Word2Vec-led term with strong contextual support (*bloqueio*), a TF-IDF-led term with strong contextual support (*salário*), a theory seed whose contextual rank rises despite modest static drift (*reforma*), and a stable-control diagnostic term (*trabalho*) that reveals where contextual leakage remains.

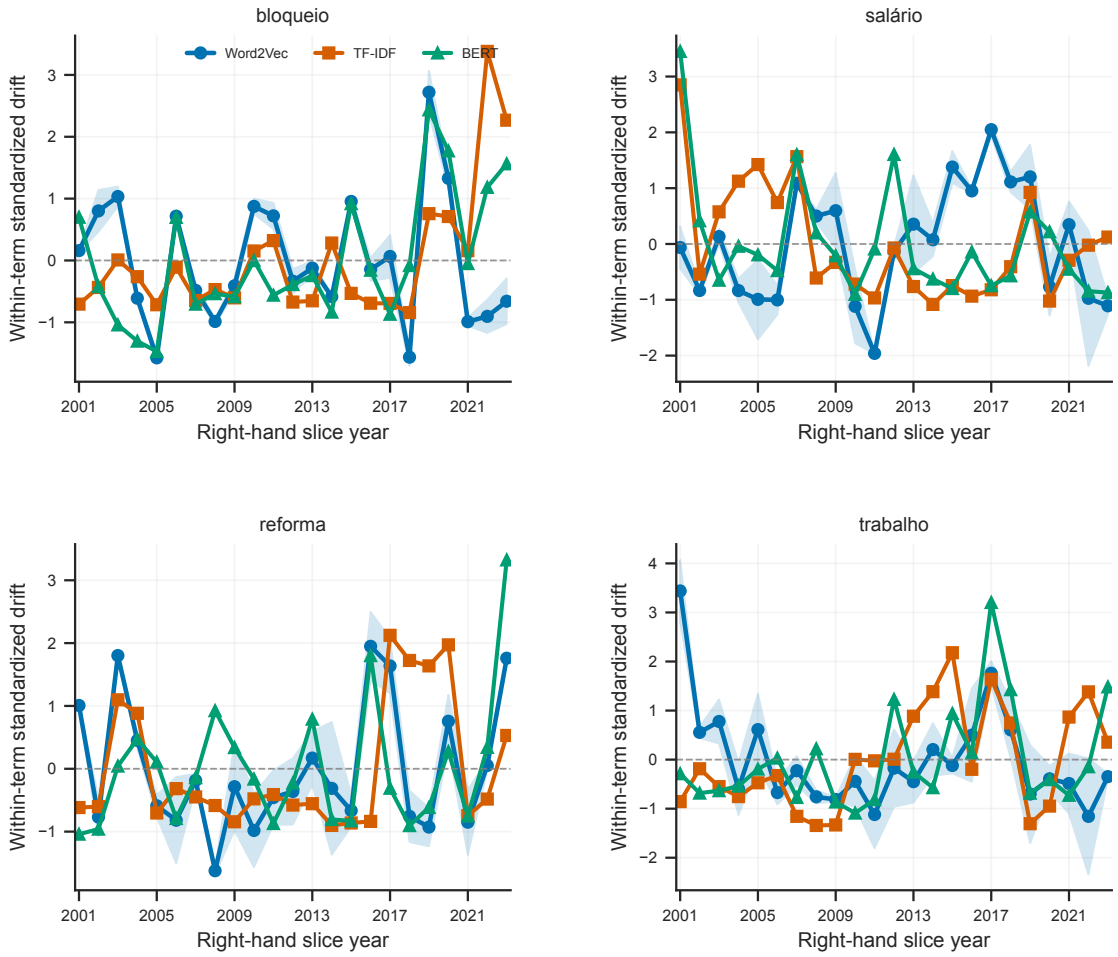

Figure 4. Representative term trajectories for *bloqueio* (top left), *salário* (top right), *reforma* (bottom left), and *trabalho* (bottom right). Scores are standardized within term and method to emphasize temporal shape rather than absolute scale; legend in the top-left panel applies throughout. Word2Vec ribbons denote replicate variability.

The disagreement itself is substantively informative, because each method foregrounds a different facet of lexical change. TF-IDF is most sensitive to lexical-profile reweightings, so a term whose salience rises in yearly agendas, policy disputes, or formulaic framing can register as drifting even if its local semantic neighborhood remains comparatively stable. Word2Vec, by contrast, captures persistent neighborhood reorganization across slices. Contextual BERT occupies an intermediate position and can support either a TF-IDF-led or a Word2Vec-led candidate through its token-level representations, although the stable-control leakage analysis shows that greater context sensitivity does not automatically produce a cleaner unrestricted ranking.

The term-level evidence supports this interpretation. *Salário* behaves like a lexical-profile case with strong contextual support, suggesting that changes in agenda prominence and usage framing are both at work. *Bloqueio* looks more like a neighborhood-displacement case in which the contextual layer aligns with the embedding-led signal. *Reforma* illustrates a complementary pattern, since as a theory seed it displays contextual differentiation despite

ranking modestly in the cheaper methods. The shared panel includes theory seeds and stable controls alongside method-led drift terms to make such distinctions recoverable.

Table 2. Method scope and relative computational cost. The cost column reports order-of-magnitude class and approximate multiplier relative to TF-IDF, measured on the same single-machine run; absolute wall-clock times will vary with hardware.

Method	Signal focus	Scope	Relative cost
TF-IDF	Adjacent-slice salience shift	3,221 lemmas	seconds ($1\times$)
Word2Vec	Aligned neighborhood displacement	3,221 lemmas	minutes ($\sim 56\times$)
Contextual BERT	Usage divergence from sampled contexts	55 lemmas / 82,461 occurrences	hours ($\sim 600\times$)

Table 2 quantifies the cost gap. TF-IDF runs in seconds, Word2Vec in minutes at corpus scale, and contextual BERT is roughly two orders of magnitude more expensive even after restriction to the 55-lemma panel. Given the modest rank correlations reported above, the current evidence does not support replacing the cheaper methods with BERT but instead favours using contextual modeling selectively as an adjudication layer when the disagreement is substantively interesting.

7. Conclusion

Returning to the research questions posed in the introduction, the evidence supports three answers. First, regarding the strength of inter-method agreement, TF-IDF and Word2Vec disagree strongly on the shared panel (Spearman $\rho = -0.54$, zero top-15 overlap), and contextual BERT remains only weakly correlated with either baseline ($\rho = 0.21$ with Word2Vec, $\rho = 0.12$ with TF-IDF), so the three families do not converge on a single ranking. Second, regarding method-specific candidates, all 30 drift terms nominated by the two cheaper methods are exclusive to one baseline; BERT partially bridges this split by placing candidates from both camps among its top-ranked terms, but it also retains some stable controls in high positions, so its raw ranking should not be treated as an unrestricted candidate list. Third, regarding the cost–benefit of the contextual stage, BERT provides useful adjudication when inter-method disagreement is substantively interesting, as illustrated by terms such as *bloqueio*, *salário*, and *reforma*. At roughly $600\times$ the cost of TF-IDF, though, the current evidence favors using it selectively rather than as a replacement for the cheaper methods. Taken together, these results frame disagreement itself as part of the contribution, suggesting that the three method families are sensitive to different mixtures of lexical salience, neighborhood displacement, and usage-level variation in Brazilian Portuguese political discourse. Because the comparative framework (Section 3) specifies roles and interfaces rather than particular models, it can be instantiated with alternative static or contextual embeddings, applied to other languages, or extended with additional tiers.

8. Limitations

The primary evaluative limitation is that this paper does not claim external semantic ground truth for BrPoliCorpus floor, so agreement and disagreement are analyzed as comparative

evidence rather than as proof that one detector is correct. A second, methodological limitation is that the raw contextual ranking still admits some stable controls near the top, even though the filtered contextual list is useful for interpretation; contextual sensitivity does not automatically yield a cleaner unrestricted candidate list. The choice of contextual aggregation strategy also merits discussion. We report centroid-based drift as the primary contextual metric and verify it against APD [Giulianelli et al. 2020, Cassotti et al. 2024] (Section 5); the strong correlation and the centroid metric's tighter bucket separation support this choice, though APD captures complementary embedding-cloud variance. Sense-based clustering methods [Montariol et al. 2021] were not attempted here. Finally, yearly slicing is appropriate for a first comparative baseline but compresses within-year shocks, campaign cycles, and short-lived institutional disputes that may matter for political language.

Future work could address the evaluative gap through external annotation or noisy supervision from labeled parliamentary datasets, though such resources still require careful deduplication before serving as reliable comparison targets. On the modeling side, adopting task-specialized models such as XL-LEXEME [Cassotti et al. 2024], exploring newer divergence metrics [Rachinskiy and Schlechtweg 2025], and applying orthographic-bias mitigation through subword masking [Laicher et al. 2021] would bring the contextual stage closer to current best practices. A symbolic companion analysis using lexical complexity or readability features could also help separate lexical-semantic drift from rhetorical or stylistic change.

9. Ethics Statement

The study analyzes public institutional political speech and reports only aggregate corpus statistics, ranked lemmas, and usage-level comparative signals. We do not profile individual speakers or infer private attributes. Even so, political corpora encode unequal representation and occasionally harmful rhetoric, so drift scores should not be read as normative judgments, evidence of speaker intent, or standalone claims about social meaning. All outputs are exploratory comparative signals whose interpretation depends on corpus composition, preprocessing choices, and the distinction between lexical salience, contextual variation, and broader discourse change.

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